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(54) **LIQUID FUEL ON-BOARD
DESULFURIZATION SYSTEMS AND
METHODS**

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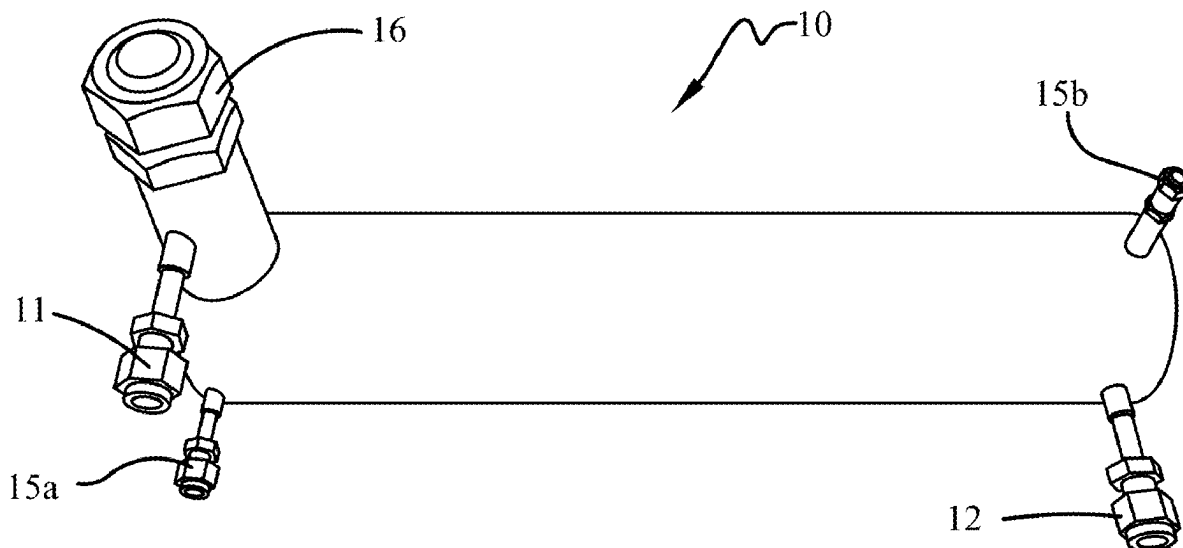
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(57) **ABSTRACT**

Systems and methods for removing sulfur from a liquid fuel feedstock, the system including a single-piece desulfurization reactor for producing a desulfurized liquid fuel stream, the single-piece desulfurization reactor including a fuel inlet, a fuel outlet, a helical fuel path in fluid communication with the fuel inlet and the fuel outlet, wherein the helical fuel path includes a metal oxide sorbent material, a reactor core formed within the single piece desulfurization reactor configured to externally receive at least one heating element, wherein the helical fuel path is formed around the reactor core, and a heat exchanger for cooling the desulfurized liquid fuel stream in fluid communication with the fuel outlet.



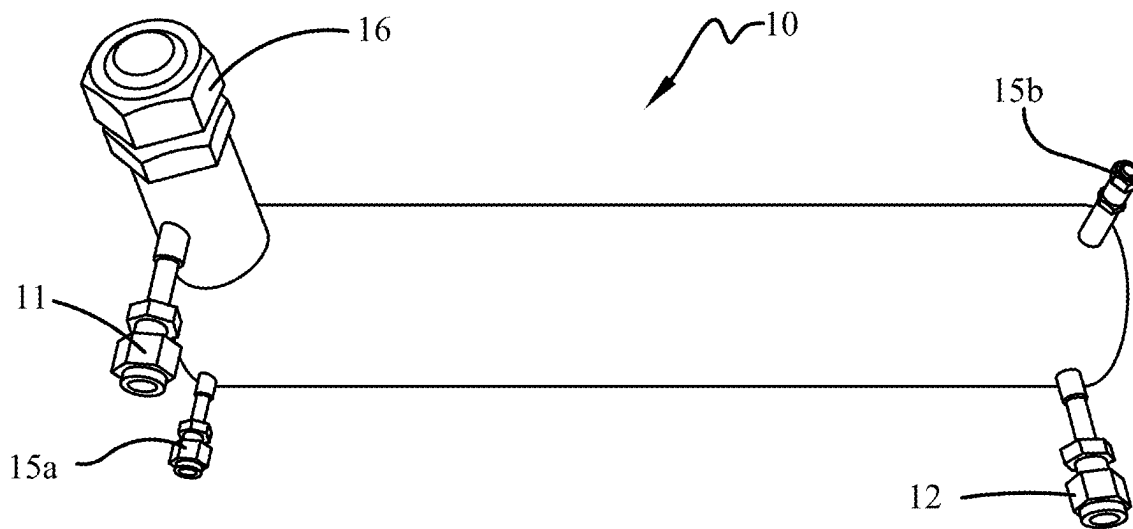


Figure 1

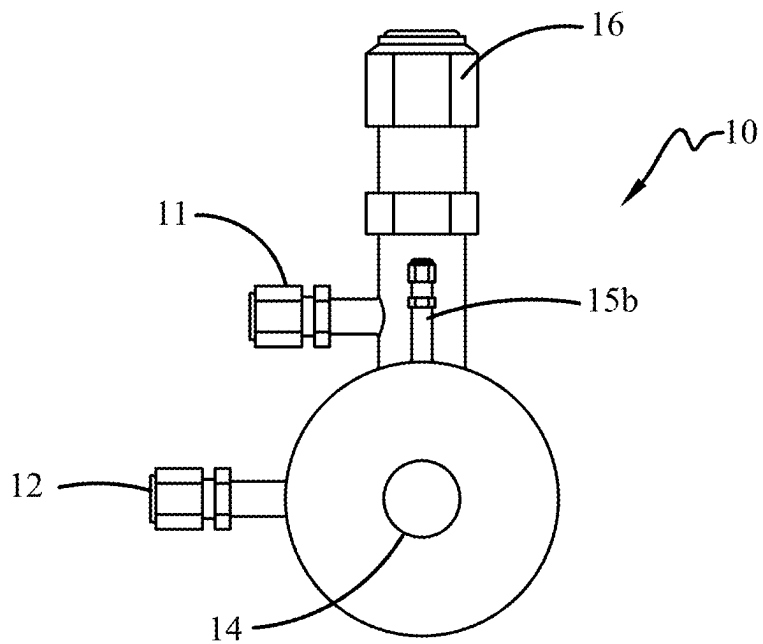


Figure 2

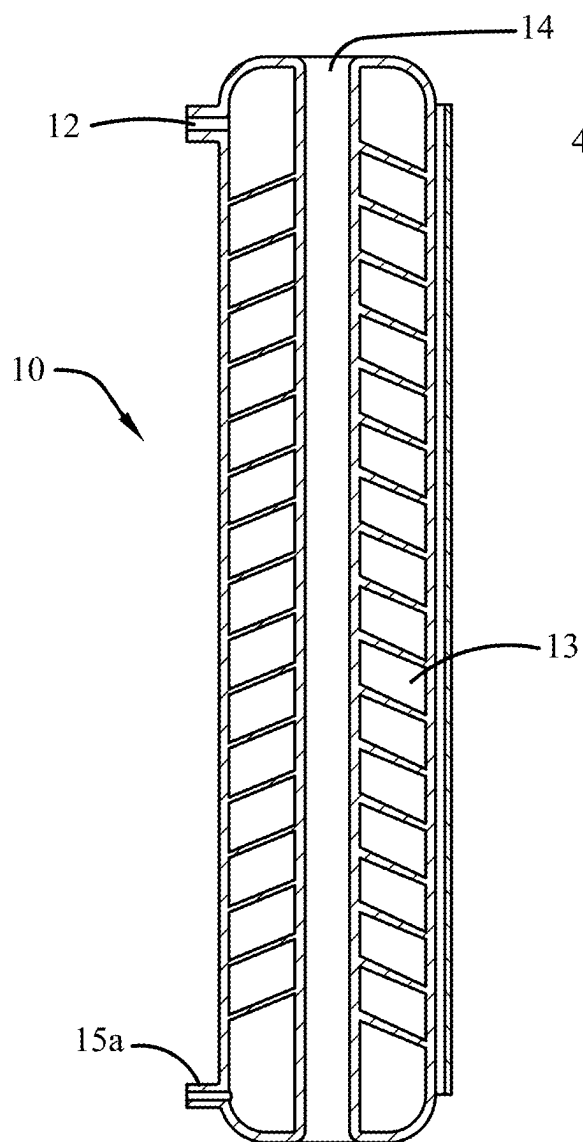


Figure 3

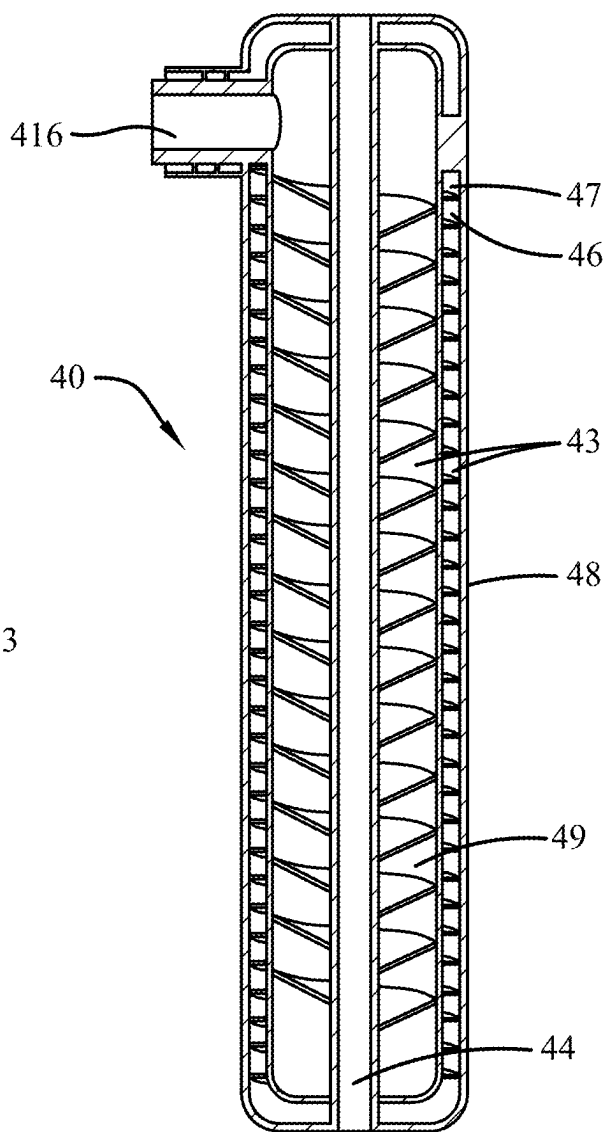


Figure 4

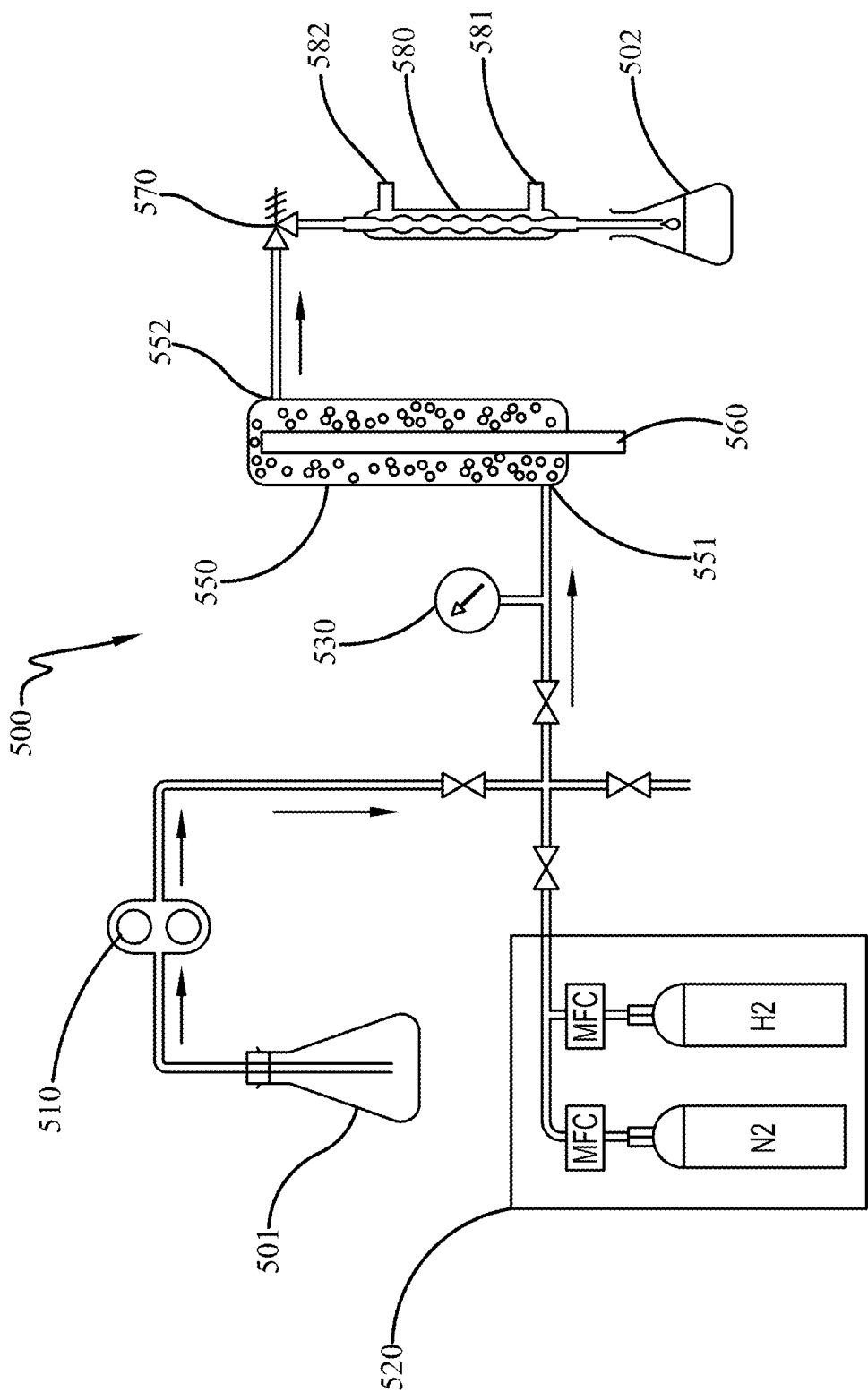


Figure 5

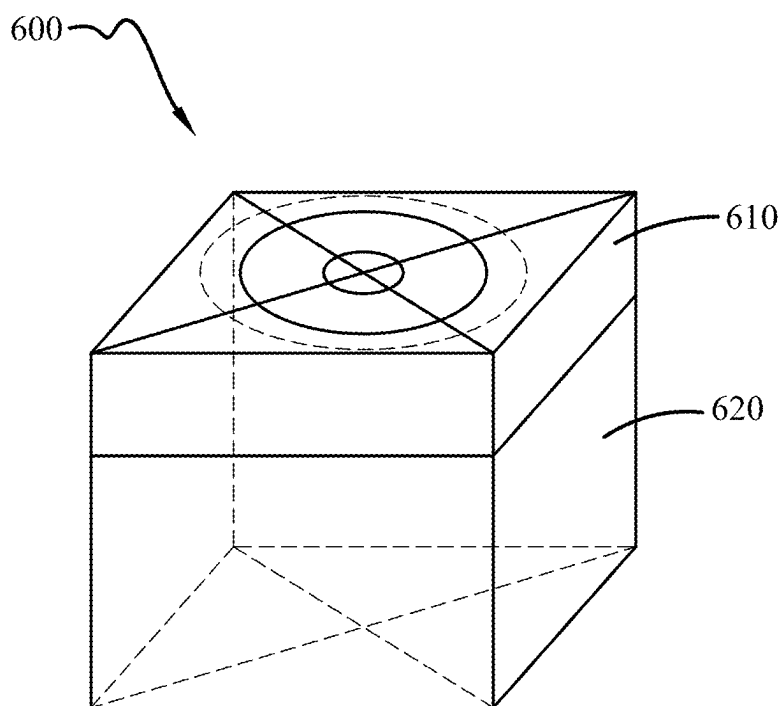


Figure 6

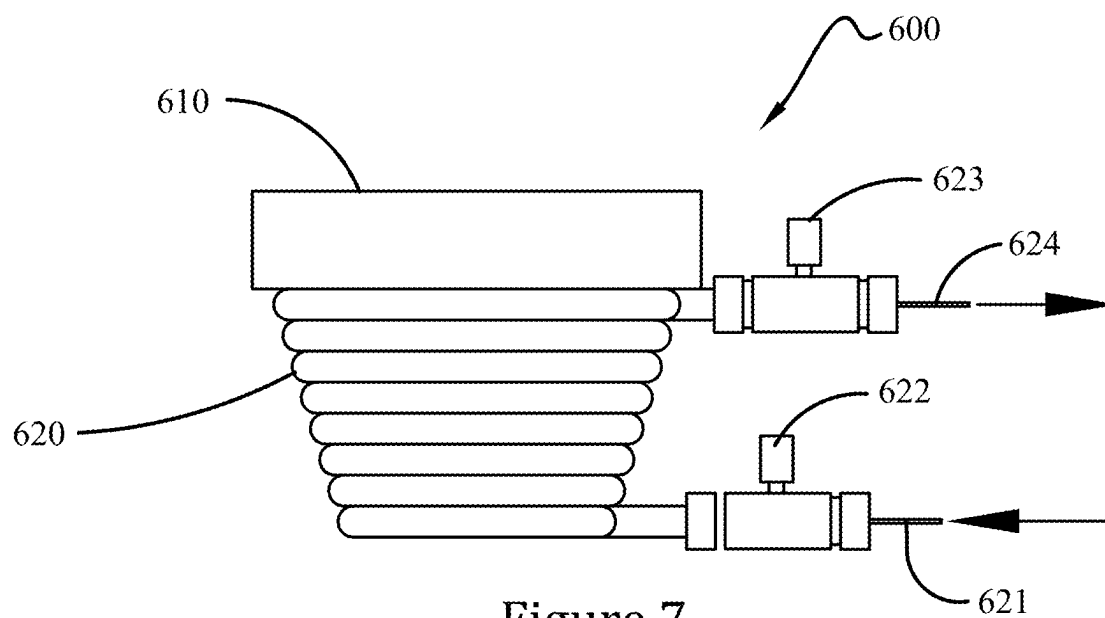


Figure 7

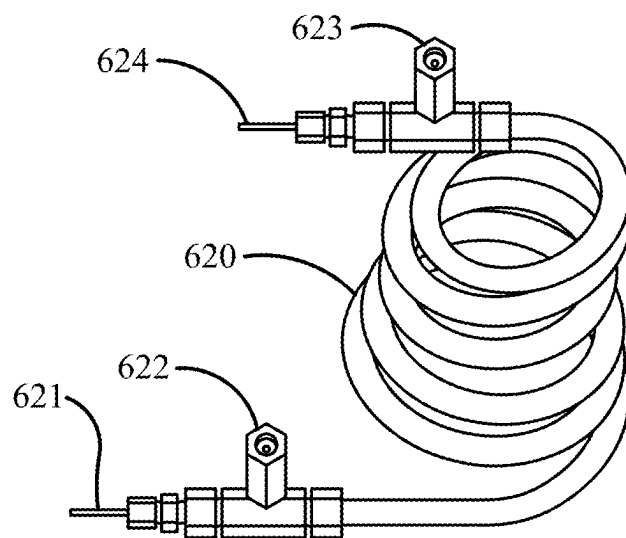


Figure 8

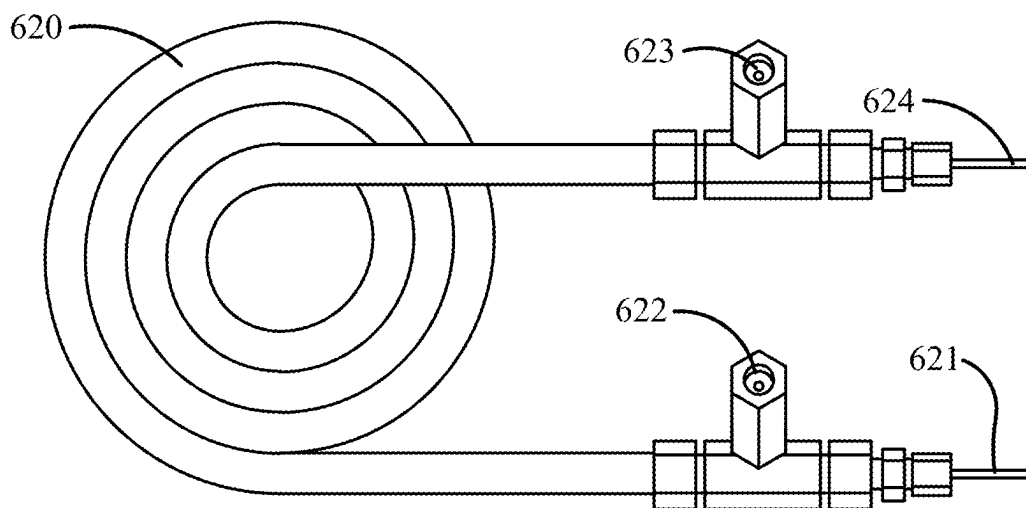


Figure 9

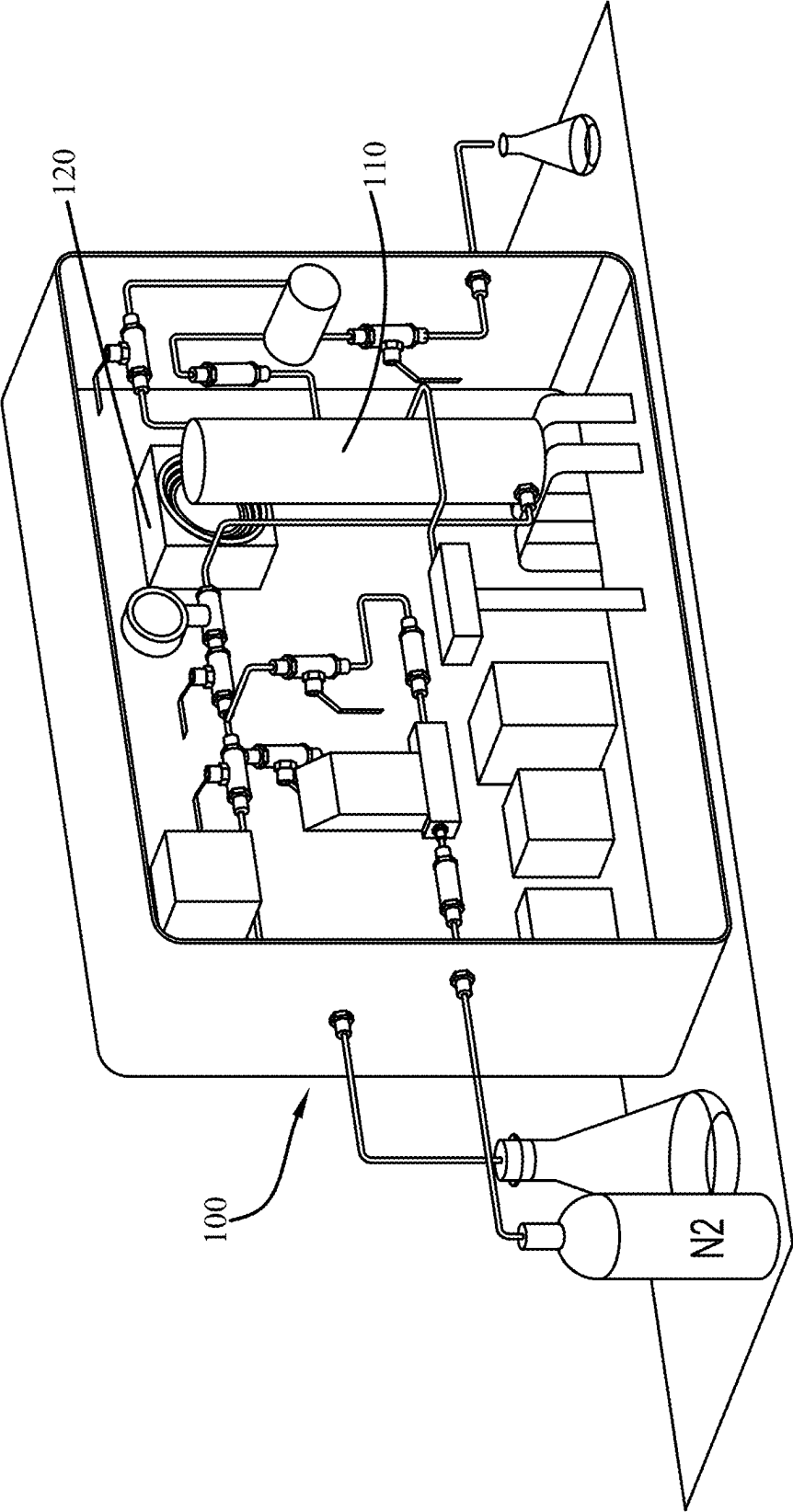


Figure 10

LIQUID FUEL ON-BOARD DESULFURIZATION SYSTEMS AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the benefit of U.S. Provisional Patent Application No. 63/513,196, filed on Jul. 12, 2023, which is incorporated herein in its entirety.

[0002] STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0003] This invention was made with government support under FA864920P0387 and FA864920P0977 awarded by the United States Air Force STTR Phase I and Phase II, respectively. The government has certain rights in the invention.

FIELD OF THE INVENTION

[0004] The present invention relates to a miniaturized desulfurization systems and methods to effectively remove sulfur from liquid fuels. The present invention specifically relates to the treatment of diesel, JP-8, and other jet fuels in an onsite, on-board or on-demand operation.

BACKGROUND OF THE INVENTION

[0005] Fuel cells are one of the most efficient electrical power generation technologies available. Among various types of fuel cells, solid oxide fuel cells (SOFCs) offer promising features for clean and efficient power generation from a variety of fuels. SOFCs operate at high temperatures which allows for the capability of using liquid hydrocarbon fuels, including jet fuel amongst others.

[0006] Jet fuel presents a promising fuel for solid oxide fuel cell power systems due to its high energy density and availability. Jet fuel is a widely used logistic fuel in both commercial and military applications, such as aircraft and ground equipment. Of the many variants of jet fuel, Jet A and Jet A-1 are the types most often used in commercial aviation. Military grade jet fuels are classified based on Jet Propellant (JP) numbers and the commonly used grades include JP-8 and JP-5. Notably, some military jet fuels are identical to commercial jet fuels except for the amounts of certain additives. For example, the composition of JP-8 is similar to that of Jet A but the former contains additional additives for corrosion inhibition and static charge dissipation. In addition to aviation applications, the military jet fuels are also used for fueling the ground vehicles and equipment in battlefields. The wide availability and high energy density of jet fuel makes it an attractive fuel source for SOFCs, especially in transport and portable applications.

[0007] A major challenge for readily using the jet fuel in fuel cell systems is its sulfur content. At present, fuel reforming catalysts and SOFC anodes are easily fouled/poisoned by sulfur compounds present in liquid fuels. The maximum sulfur content in jet fuels could be as high as 3000 ppm_w (parts per million by weight) based on current regulatory standards although typical sulfur contents are found to vary between 300 and 700 ppm_w. There are various organic sulfur compounds in jet fuels with benzothiophene and methylated benzothiophene being the two main species. These sulfur compounds can deactivate the catalysts that are used for on-board fuel reforming in fuel cell systems. Moreover, the organic sulfur compounds are converted to

H₂S during fuel reforming. H₂S is detrimental to SOFC anodes because of its poisoning effects even at concentrations of a few ppm. Different possible mechanisms have been reported for the sulfur poisoning of state-of-the-art Ni-based SOFC anodes. Sulfur adsorption on the anode surface is generally believed to be the major cause of SOFC performance degradation at low H₂S concentrations. Sulfur adsorption may take place physically (adsorption of molecular H₂S on Ni) or chemically (adsorption of dissociated sulfur atom on Ni). In either case, the adsorbed sulfur blocks the electrochemical reaction sites of the anode, thus degrading the SOFC performance. For higher level of H₂S concentrations (≥100 ppm_w), the formation of nickel sulfide compounds has been reported to be responsible for anode deactivation. As such, the effective removal of sulfur from liquid fuels is required prior to supplying the liquid fuel to SOFC power systems.

[0008] A number of technologies have been developed to remove sulfur from liquid hydrocarbon fuels, or reformat streams, for fuel cell utilization. Broadly, these technologies are based on two types of desulfurization methods: hydrodesulfurization (HDS) and adsorptive desulfurization (ADS). In an HDS reactor, organic sulfur compounds catalytically react with hydrogen to form hydrogen sulfide, which is subsequently converted to zinc sulfide over a bed of nickel oxide. The typical catalysts for HDS include supported nickel-molybdenum oxide and cobalt-molybdenum oxide. Although HDS is an effective method for removing lighter sulfur compounds, its performance is poor towards larger sulfur compounds such as methylated benzothiophenes and dibenzothiophenes. Moreover, the HDS technology requires a continuous supply of hydrogen gas. This is a particular concern for internal-reforming SOFCs because they do not have a hydrogen-rich stream that can be fed to the HDS reactor. As an alternative to HDS, there has been growing attention to ADS, which uses adsorbents to selectively adsorb the organic sulfur compounds. Various adsorbents, including activated carbon, silica, metal oxides, zeolites, aluminosilicates, and metal-organic frameworks, have been utilized for the desulfurization of liquid hydrocarbon fuels. In addition to being effective in trapping larger sulfur compounds, ADS does not require hydrogen as a reactant. Another advantage of the ADS process is that it can be carried out at relatively low temperature and pressure conditions, thus further simplifying the system design. Previous designs of ADS processes are included in U.S. Pat. No. 9,238,781 which is incorporated herein by reference to its entirety.

[0009] Accordingly, there is a need for desulfurization systems and methods with reduced complexity, size, and weight while enabling operation in all directions and orientations.

SUMMARY OF THE INVENTION

[0010] One or more embodiments of the present invention provide a single-piece desulfurization reactor comprising a fuel inlet, a fuel outlet, a helical fuel path in fluid communication with the fuel inlet and the fuel outlet, wherein the helical fuel path includes a metal oxide sorbent material, a reactor core formed within the single piece desulfurization reactor configured to externally receive a heating element, wherein the helical fuel path is formed around the reactor core.

[0011] Another embodiment of the present invention provides a single-piece desulfurization reactor as in any embodiment above, wherein the helical fuel path comprises a single-helical reactor portion including the metal oxide sorbent material formed around the reactor core, and one or more heat exchange portions comprising a double-helical portion formed around the single-helical portion in fluid communication with the fuel inlet, the single-helical reactor portion, and the fuel outlet.

[0012] Another embodiment of the present invention provides a single-piece desulfurization reactor as in any embodiment above, wherein a liquid fuel feedstock cannot bypass a portion of the helical fuel path within the single-piece desulfurization reactor.

[0013] Another embodiment of the present invention provides a single-piece desulfurization reactor as in any embodiment above, wherein the metal oxide sorbent material comprises nickel oxide.

[0014] Another embodiment of the present invention provides a single-piece desulfurization reactor as in any embodiment above, wherein the single-piece desulfurization reactor produces a treated output including 15 ppm_w sulfur or less.

[0015] Another embodiment of the present invention provides a single-piece desulfurization reactor as in any embodiment above,

[0016] Yet other embodiments of the present invention provide a system for removing sulfur from a liquid fuel feedstock, the system comprising a single-piece desulfurization reactor for producing a desulfurized liquid fuel stream, the single-piece desulfurization reactor comprising: a fuel inlet, a fuel outlet, a helical fuel path in fluid communication with the fuel inlet and the fuel outlet, wherein the helical fuel path includes a metal oxide sorbent material, a reactor core formed within the single piece desulfurization reactor configured to externally receive at least one heating element, wherein the helical fuel path is formed around the reactor core; a heat exchanger for cooling the desulfurized liquid fuel stream in fluid communication with the fuel outlet.

[0017] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the helical fuel path comprises a single-helical reactor portion including the metal oxide sorbent material formed around the reactor core, and one or more heat exchange portions comprising a double-helical portion formed around the single-helical portion in fluid communication with the fuel inlet, the single-helical reactor portion, and the fuel outlet.

[0018] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the heat exchanger is a condenser, wherein the condenser removes heat from the desulfurized liquid fuel stream via heat exchange with a water stream, wherein after removing heat from the desulfurized liquid fuel stream a warmed water stream is produced.

[0019] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the heat exchanger comprises a fan and a tube-in-tube heat exchanger.

[0020] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel

feedstock as in any embodiment above, wherein the tube-in-tube heat exchanger comprises an interior tube positioned within an exterior tube, wherein the interior tube includes an interior inlet in fluid communication with the liquid fuel feedstock and an interior outlet in fluid communication with the fuel inlet of the reactor, wherein the exterior tube includes an exterior inlet in fluid communication with the fuel outlet of the reactor, and an exterior outlet in fluid communication with a desulfurized fuel reservoir.

[0021] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the tube-in-tube heat exchanger is configured for counter-current flow with the interior inlet and exterior outlet being collocated at a first end of the tube-in-tube heat exchanger and the interior outlet and exterior inlet being collocated at a second end of the tube-in-tube heat exchanger.

[0022] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the system excludes any further heating source apart from the at least one heating element in the reactor core.

[0023] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the tube-in-tube heat exchanger further includes an external fan for moving air across the tube-in-tube heat exchanger.

[0024] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, further comprising an activation and regeneration system comprising a source of nitrogen gas, hydrogen gas, and mixtures thereof.

[0025] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein one or more additional single-piece desulfurization reactors in fluid communication with the liquid fuel feedstock.

[0026] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the single-piece desulfurization reactors and the one or more additional single-piece desulfurization reactors are independently controlled to provide continuously produce the desulfurized fuel stream.

[0027] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the single-piece desulfurization reactor is configured for quick-replacement by one of the at least one or more additional single-piece desulfurization reactors to maintain continuous production of the desulfurized liquid fuel stream.

[0028] Another embodiment of the present invention provides a system for removing sulfur from a liquid fuel feedstock as in any embodiment above, wherein the liquid fuel feedstock comprises JP-8 fuel.

[0029] Yet other embodiments of the present invention provide method of removing sulfur from a liquid fuel feedstock, the method comprising providing the liquid fuel feedstock to a single-piece desulfurization reactor, wherein the liquid fuel feedstock is in fluid communication with the fuel inlet; flowing the liquid fuel feed stock through the helical fuel path to contact the metal oxide sorbent material; controlling a temperature of a heating element located with the reactor core; controlling a pressure within the single-

piece desulfurization reactor; and collecting a desulfurized liquid fuel stream produced by the single-piece desulfurization reactor from the fuel outlet.

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1 is a perspective view of a single-piece desulfurization reactor according to an embodiment of the present invention.

[0031] FIG. 2 is a perspective view of a single-piece desulfurization reactor according to an embodiment of the present invention.

[0032] FIG. 3 is a sectional view of a single-piece desulfurization reactor according to an embodiment of the present invention.

[0033] FIG. 4 is a sectional view of a single-piece desulfurization reactor according to an embodiment of the present invention.

[0034] FIG. 5 is a schematic view of a system for removing sulfur from a liquid fuel feedstock according to an embodiment of the present invention.

[0035] FIG. 6 is a perspective view of a heat exchanger according to an embodiment of the present invention.

[0036] FIG. 7 is a side view of a heat exchanger according to an embodiment of the present invention.

[0037] FIG. 8 is a perspective view of a tube-in-tube heat exchanger according to an embodiment of the present invention.

[0038] FIG. 9 is a perspective view of a tube-in-tube heat exchanger according to an embodiment of the present invention.

[0039] FIG. 10 is a perspective view of a system for removing sulfur from a liquid fuel feedstock according to an embodiment of the present invention.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

Introduction

[0040] Embodiments of the invention are based, at least in part, on the discovery of reactors, systems and methods for removing sulfur from liquid fuels. In particular, the present invention is directed towards a single-piece adsorptive desulfurization reactor having a fuel inlet, fuel outlet, and a helical fuel path including a metal oxide sorbent material. Embodiments of desulfurization reactors according to the present invention advantageously reduce the weight, size, and complexity of previous desulfurization processes and reactors. In particular, embodiments of the present invention provide a leak-free operation which enhances desulfurization performance and increases suitability for diverse operating environments. Further advantages of the single-piece construction include enhanced portability and miniaturization of desulfurization reactors.

[0041] Embodiments of the present invention are suitable for use in any application where sulfur needs to be removed from a liquid fuel to prevent sulfur corrosion or sulfur poisoning of another process or application. Examples of such applications include reducing equipment cost due to sulfur corrosion by enhancing the longevity of metal fuel containers and fuel tanks by storing desulfurized liquid fuels treated according to the present invention. Other exemplary applications include the direct use of desulfurized fuel for

power generation in fuel cells to generate power, as fuel for vehicles and for reducing sulfur oxide (SO_x) combustion byproducts.

[0042] Embodiments of the present invention may be sized for a variety of throughput requirements. In particular, in an application for processing diesel and other liquid fuels systems and methods according to the present invention may treat up to 500 gallons per day of liquid fuel. In other embodiments, systems and methods according to the present invention may treat enough liquid fuel to generate 2 kW via a fuel cell for 100 hours.

[0043] Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

Single-Piece Desulfurization Reactor

[0044] Referring to FIGS. 1-3, a single-piece desulfurization reactor 10 in accordance with an embodiment of the present invention is illustrated. Reactor 10 has a generally cylindrical shape and includes fuel inlet 11, fuel outlet 12, helical fuel path 13, reactor core 14, and thermocouple ports 15a, 15b. Fuel inlet 11 and fuel outlet 12 provide openings into reactor 10 and are in fluid communication with helical fuel path 13. In some embodiments, helical fuel path 13 includes a metal oxide sorbent material dispersed therein for removing sulfur from a liquid fuel flowing within helical fuel path 13. The sorbent material may be added or removed to helical fuel path 13 through sorbent port 16.

[0045] In these and other embodiments the fuel inlet, the fuel outlet, the sorbent port, and other features may further include tubing and piping that is joined to the external surface of the reactor. Such tubing and piping may further include a variety of connections for providing a leak-free connection to the reactor. Such connections may include fittings provided by Swagelok and others.

[0046] As shown in FIGS. 1-3, single-piece desulfurization reactor 10 is formed as a unitary body which prevents leaks from occurring between individual portions of helical fuel path 13 as the walls of helical fuel path 13 are joined together without any gaps.

[0047] As shown in FIG. 4, single-piece desulfurization reactor 40 in accordance with the present invention is illustrated. Reactor 40 has a generally cylindrical shape and includes a fuel inlet, a fuel outlet, helical fuel path 43, including single-helical reactor portion 49 and double-helical portion 48, which includes inlet side 46 and outlet side 47, and reactor core 44. The inlet and the fuel outlet provide openings into reactor 40 and are in fluid communication with helical fuel path 43. Fuel is introduced to helical fuel path 43 via the fuel inlet and exits helical fuel path 43 via the fuel outlet. Specifically, the fuel inlet is in fluid communication with inlet side 46 of double-helical portion 48 of helical fuel path 43. The fuel outlet is in fluid communication with outlet side 47 of double-helical portion 48 of

helical fuel path 43. Further, inlet side 46 is in fluid communication with single-helical reactor portion 49 which is in fluid communication with outlet side 47. As such, fuel flows into reactor 40 through the fuel inlet into helical path 43 first, via inlet side 46 of double-helical portion 48 positioned around single-helical reactor portion 49, then into single-helical reactor portion 49, then exiting single-helical reactor portion 49 into outlet side 47 of double-helical portion 48 before exiting reactor 40 via the fuel outlet. Accordingly, within double-helical portion 48, heat exchange can occur between inlet side 46 and outlet side 47 to facilitate an increase in temperature of the liquid fuel flowing through inlet side 46, and a decrease in temperature of desulfurized liquid fuel flowing through outlet side 47. Fuel flowing through inlet side 46 is additionally warmed by heat transfer through the wall between single-helical reactor portion 49 and double-helical portion 48. In some embodiments, single-helical reactor portion 49 includes a metal oxide sorbent material dispersed therein for removing sulfur from a liquid fuel flowing within single-helical reactor portion 49. In these and other embodiments, the sorbent material may be added or removed to helical fuel path 43 through sorbent port 416. In some embodiments, more than one double-helical portion may be present to provide further residence time to cause heat transfer between the incoming fuel to be treated and treated fuel exiting the reactor.

[0048] Reactor cores 14, 44 are formed as cylindrical voids in the center of reactors 10, 40. Reactor cores 14, 44 are configured to house one or more heating elements (not shown) to thereby provide heat to the reactor and the liquid fuel contained therein during a desulfurization process. Multiple thermocouples may be employed to monitor the temperature at various points of reactors 10, 40. For example, thermocouples may be placed within thermocouple ports 15a, 15b to monitor and/or control the temperature of the liquid fuel at fuel inlet 11 and fuel outlet 12, as well as helical fuel path 13 via changing. Monitoring of the temperature at various points of reactor 10 allows for control of the heating elements to maintain process stability. Reactor cores 14, 44 may accommodate a broad range of heaters for providing heat to the core of the reactor. For example, reactor cores 14, 44 may receive an electrical heating element which converts electrical energy into heat energy to warm liquid fuel within reactors 10, 40. Such electrical heating elements may also be used to provide heat to reactors 10, 40 during regeneration processes of the sorbent material as well.

[0049] Reactors 10, 40 may also be heated by one or more external heating elements. Such external heating elements may include tape style electrical heating elements wrapped around the exterior of reactors 10, 40. In some embodiments, such as for example in reactor 40, the presence of double-helical portion 48, reduces the temperature gradient across the radius of reactor 40 such that no external heating is required to maintain operational performance.

[0050] As discussed above, helical fuel paths 13, 43 include one or more sorbent materials (not shown) for desulfurizing the liquid fuel that passes therethrough. For example, a regenerable metal oxide sorbent material can be utilized. Regenerable desulfurization agents are capable of removing sulfur from refractory organosulfur compounds. Such desulfurization agents can remove sulfur from refractory organosulfur compounds at moderate temperature and at moderate pressure in the absence of hydrogen and the

desulfurization agent can be fully regenerated by exposing it to air at a temperature that is substantially the same as that at which the desulfurization agent removes sulfur from the fuel. While the temperature may not remain precisely static, no active steps are taken to raise or lower the temperature. Without wishing to be bound by any particular theory, it appears that metal or metal oxide species are converted to metal sulfides during desulfurization, and that the metal sulfides are converted back to metal oxide species during the regeneration process. In the presence of a reducing, sulfur-containing gas, the Gibbs free energy of the chemical system is minimized if the metal or metal oxide converts to the corresponding metal sulfide. In the presence of an oxidizing gas, the Gibbs free energy of the chemical system is minimized if the metal sulfide transforms to the corresponding metal oxide.

[0051] In accordance with the present invention, one suitable metal oxide sorbent material includes nickel oxide. The metal oxide sorbent material can be activated before desulfurizing liquid fuel. The activation can be accomplished by flowing mixture of nitrogen and hydrogen (for example, at N_2/H_2 98/2 ratio) at high temperature (400° C. for example) at ambient pressure for a period of time (8 hours or greater for example). The main purpose for the activation is to remove oxygen gas from within the reactor and from the absorbent material.

[0052] Some embodiments of single-piece desulfurization reactors according to the present invention produce desulfurized liquid fuel including 15 ppm_w sulfur or less. In other embodiments, single-piece desulfurization reactors produce desulfurized liquid fuel including 10 ppm_w sulfur or less. In other embodiments, single-piece desulfurization reactors produce desulfurized liquid fuel including 5 ppm_w sulfur or less. In other embodiments, single-piece desulfurization reactors produce desulfurized liquid fuel including 1 ppm_w sulfur or less.

[0053] In some embodiments, the fuel outlet including desulfurized liquid fuel is in fluid communication with a fuel cell system. In these and other embodiments, the fuel outlet is in liquid communication with a fuel cell manifold. In these and other embodiments, the fuel outlet is in fluid communication with a fuel reformer in fluid communication with a fuel cell system.

[0054] Single-piece desulfurization reactors according to the present invention may be formed by a variety of manufacturing techniques and processes. In some embodiments, single-piece desulfurization reactors according to the present invention are formed by additive manufacturing processes including 3D printing processes.

[0055] Advantages of single-piece desulfurization reactors according to the present invention further include decreased weight and increased sorbent bed volume for a given reactor volume relative to conventional techniques. For example, a single-piece desulfurization reactor according to some embodiments of the present invention has a mass reduction of 40% or greater relative to a conventional reactor of the same total volume. Likewise, the same single-piece desulfurization reactor according to an embodiment of the present invention has a sorbent bed volume (i.e. the amount volume of sorbent in the fuel path) that is 50% or greater relative to a conventional reactor of the same total volume.

[0056] Embodiments of single-piece desulfurization reactors according to the present invention are readily config-

unable to meet target performance requirements relating to sulfur content, flow rate and desulfurized fuel volume, and regeneration cycling. For example, some embodiments of single-piece desulfurization reactors according to the present invention are sized to treat 1 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 2 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 3 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 4 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 5 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 10 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 15 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 20 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 25 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 30 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 35 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 40 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 45 mL/min of liquid fuel. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 50 mL/min of liquid fuel.

[0057] In some embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 200 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 400 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 600 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 800 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 1000 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 1200 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 1400 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 1600 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the present invention are sized to treat 1800 mL or more of liquid fuel before requiring regeneration. In other embodiments, single-piece desulfurization reactors according to the pres-

ent invention are sized to treat 2000 mL or more of liquid fuel before requiring regeneration.

[0058] The person of ordinary skill in the art understands that the regeneration schedule is dependent upon the sulfur content in the liquid fuel being treated, the performance of the selected sorbent, and the desired level of sulfur remaining in the treated fuel. As such, the throughput capacity of reactors according to the present invention may be greatly increased.

[0059] The person of ordinary skill in the art understands that performance of single-piece desulfurization reactors according to the present invention is a function of the physical dimensions of the reactor, the operating conditions, and the volume of sorbent material.

[0060] Embodiments of single-piece desulfurization reactors according to the present invention may have a variety of thicknesses of different structural portions of the reactor. For example, the walls, which includes the surfaces that define a volume of the helical fuel path may have a variety of thicknesses. As a result of the single-piece construction, reactors according to the present invention may have thinner walls than reactors formed by conventional techniques, while being able to withstand operational conditions of desulfurization and regeneration processes. In some embodiments, single-piece desulfurization reactors according to the present invention may have a wall thickness of 0.5 mm or greater. In other embodiments, a wall thickness of 1.0 mm or greater. In other embodiments, 1.5 mm or greater. In other embodiments, a wall thickness of 2.0 mm or greater. In other embodiments, 2.5 mm or greater. The person of ordinary skill in the art further understands that the thickness of the reactor is influenced by the size and dimensions of the reactor and fuel capacity in order to support the operating conditions (i.e. pressure/temperature) required.

[0061] Embodiments of single-piece desulfurization reactors according to the present invention may be constructed to various sizes. Additive manufacturing processes provide for a large variety of geometries and sizes. The person of ordinary skill in the art understands that the capacity of the reactor, including the sorbent volume within the fuel path increases as reactor size increases. Accordingly, the size of single-piece desulfurization reactors can be adjusted according to any one of capacity requirements relating to overall size of the system, manufacturing capabilities, and other demands.

Desulfurization System and Methods

[0062] Referring to FIG. 5, desulfurization system 500 includes liquid fuel reservoir 501, in fluid communication with liquid fuel pump 510, in fluid communication with single-piece desulfurization reactor 550, in fluid communication with relief valve 570, in fluid communication with condenser 580, in fluid communication with desulfurized fuel reservoir 502. Desulfurization system 500 further includes activation and regeneration system 520 in fluid communication with reactor 550. Reactor 550 includes fuel inlet 551 and fuel outlet 552. Desulfurization system 500 further includes heating element 560 within reactor 550. Condenser 580 includes cold water inlet 581 and hot water outlet 582. Relief valve 570 is present to prevent over pressure events within desulfurization system 500. Desulfurization system 500 may further include valves, thermocouples, pressure gauges, such as pressure gauge 530, filters,

and other components for monitoring and controlling operation of desulfurization system 500.

[0063] In desulfurization system 500, during an activation and regeneration process, nitrogen gas, hydrogen gas, or a mixture of nitrogen and hydrogen gas is flowed from activation and regeneration system 520 to reactor 550 via fuel inlet 551 and heating element 560 heats reactor 550 to about 300 to 450° C. for a period of 6 to 24 hours at a pressure of 5 to 10 pounds per square inch (PSI). Activation processes such as those described above ensure that sorbent material within reactor 550 is activated for desulfurization processes. The temperatures, duration, and pressures of the system during activation and regeneration processes is dependent upon the size of the reactor and the type and amount of sorbent materials.

[0064] In one or more embodiments, heating element 560 includes one or more independently controllable heating elements to provide greater control over the temperature of a specific portion of reactor 550. The person of ordinary skill in the art readily appreciates that localized control of heating and increased temperature stability of the reactor allows for improved desulfurization performance and an overall improvement in process efficiency.

[0065] In desulfurization system 500, during a desulfurization process, liquid fuel to be treated by desulfurization system 500 is transported from liquid fuel reservoir 501 via liquid fuel pump 510 to reactor 550 through fuel inlet 551. The liquid fuel is then desulfurized within reactor 550 before exiting reactor 550 through fuel outlet 552. The desulfurized liquid fuel is at a high temperature and must be cooled via condenser 580 prior to storage in desulfurized fuel reservoir 502. During a desulfurization process, heating element 560 heats reactor 550 to about 150 to 300° C. In addition to heating element 560, external heating elements such as tape heaters may be present in desulfurization system 500 such as being wrapped around the exterior of reactor 550. Condenser 580 requires cooling water to provide heat exchange to cool down the desulfurized liquid fuel prior to storage in desulfurized liquid fuel reservoir 502. In order to provide cooling, condenser 580 receives cooling water through cold water inlet 581 which is then used to extract heat from the desulfurized fuel stream and the heated water is removed from condenser 580 via hot water outlet 582.

Heat Exchanger for No Condenser Operation

[0066] Referring to FIGS. 5-10, heat exchanger 600 is shown which replaces condenser 580 in desulfurization system 500 to enable desulfurization processes without the need for condenser 580. Such a modification of desulfurization system 500 greatly simplifies and reduces the complexity of desulfurization system 500 by removing the need for external cooling water and water circulation capabilities. Heat exchanger 600 provides further benefit by pre-heating liquid fuel to be treated in reactor 550 further reducing the heating requirements of heating element 560 and improving the stability and temperature uniformity of process conditions and further simplification of desulfurization system 500. The use of heat exchanger 600 further reduces the surface temperature of the reactor, which in turns avoids the need for thick thermal insulation.

[0067] Heat exchanger 600 includes fan 610 and tube-in-tube heat exchanger 620. As shown in FIGS. 6 and 7, fan 610 moves air across tube-in-tube heat exchanger 620 to provide dissipate heat to the environment. Fan 610 may include any

conventionally available fan for moving air. Fan 610 may be selected according to certain performance characteristics including power draw, variable air flow control, air flow volume, static pressure and others. Fan 610 may be of a variety of sizes according to the desired performance characteristics.

[0068] As shown in FIGS. 5-10, tube-in-tube heat exchanger 620 includes an interior tube located within an exterior tube with fittings to independently flow fluids within either the interior tube or exterior tube. As discussed above, tube-in-tube heat exchanger 620 performs a cooling of desulfurized liquid fuel that has passed through reactor 550 and warming liquid fuel before entering reactor 550. According to the embodiment shown in FIGS. 7-9, tube-in-tube heat exchanger 620 is configured in a counter-current configuration wherein liquid fuel enters tube-in-tube heat exchanger 620 at interior inlet 621 located within the diameter of exterior outlet 622 which is in fluid communication with desulfurized liquid fuel reservoir 502. Desulfurized liquid fuel enters tube-in-tube heat exchanger 620 at exterior inlet 623 which is in fluid communication with fuel outlet 552 and pre-warmed liquid fuel exits tube-in-tube heat exchanger 620 at interior outlet 624 which is in fluid communication with fuel inlet 551. Without wishing to be bound by theory it is believed that this configuration promotes a consistent temperature gradient at various points across the length of tube-in-tube heat exchanger 620 for driving heat transfer from the desulfurized liquid fuel stream to the liquid fuel stream and environment. In some embodiments, tube-in-tube heat exchanger 620, is operated in parallel configuration.

[0069] Tube-in-tube heat exchangers according to the present invention may be formed from a variety of materials and into a variety of geometries. For example, materials suitable for use in heat exchangers according to the present invention may include copper, nickel, copper and nickel alloys, and other metal alloys. Materials may be selected according to their workability in achieving a specific geometry, thermal conductivity and mechanical strength. Geometries suitable for use in the present invention include a conical geometry with spacing between coils to facilitate airflow across the exterior of the tube-in-tube heat exchanger. In such a geometry, the tubes may be described as cylindrical. Other geometries of tubes may be suitable such as diamond or square shaped tubes. Other geometries of heat exchanger may include one or more U-shaped bends.

[0070] In some embodiments, heat exchangers according to the present invention may be characterized by the degree of pre-warming of liquid fuel stream. In these embodiments, heat exchangers may warm the liquid fuel from ambient temperature to 80° C. or greater. In other embodiments, from ambient temperature to 90° C. or greater. In other embodiments, from ambient temperature to 100° C. or greater.

[0071] In some embodiments, heat exchangers according to the present invention may be characterized by the degree of cooling of the desulfurized liquid fuel stream. In these embodiments, heat exchangers may cool the liquid fuel from 200° C. or greater to 50° C. or lesser. In other embodiments, from 200° C. or greater to 45° C. or lesser. In other embodiments, from 200° C. or greater to 40° C. or lesser. In other embodiments, from 200° C. or greater to 35° C. or lesser. In other embodiments, from 200° C. or greater to 30° C. or lesser.

[0072] Referring to FIG. 10, desulfurization system 100 is shown including reactor 110 and heat exchanger 120 in a portable and packable container. Inclusion of reactor 110 and heat exchanger 120 advantageously allows for desulfurization of liquid fuel with the only requirements being a source of liquid fuel and electricity to power desulfurization system 100. As shown in FIG. 10, desulfurization system 100 is compact and storable in a container that may be man-portable, or crew-operated, and readily transported by one or more persons. Activation and regeneration capacity is added simply by including a source of materials to effect activation and regeneration. For example, in some embodiments a mixture of nitrogen and hydrogen gases may be stored separately and attached to desulfurization system 100 when activation or regeneration is required. In other embodiments, regeneration may be achieved without the need for external components.

[0073] A person of ordinary skill in the art readily appreciates that increased system throughput may be achieved by increasing the number of reactors included in the system. For example, one of ordinary skill in the art readily understands that desulfurization systems may include two or more reactors operating in parallel configuration. Parallel operation of two or more reactors provides for continuous desulfurization of liquid fuel while one or more reactors is activated or regenerated. It is further understood that the single-piece construction of reactors according to the present invention may be pre-activated and swapped into desulfurization systems via quick connections known in the art to enable continuous desulfurization operations with minimal system down time, while simultaneously simplifying the system further by removing the need for an activation and regeneration system.

Operation and Control of Desulfurization System

[0074] Embodiments of desulfurization systems according to the present invention are readily controllable via a number of programmable controllers depending upon the operative state of the system. Desulfurization systems may be broadly controlled according to flow, pressure, and temperature.

[0075] In some embodiments of desulfurization systems according to the present invention, flow control of the liquid fuel is provided via the fuel pump. In these and other embodiments, flow control of the activation and regeneration system, including nitrogen gas, hydrogen gas, and mixtures thereof is controlled by mass flow controllers. Check valves, multi-way valves, vents, and other components are used to manage flow within desulfurization systems according to the present invention. Suitable fuel pumps and mass flow controllers are known in the art. In some embodiments fuel pumps suitable for use in the present invention provide a liquid fuel flow rate of from 1 to 10 mL/min. In some embodiments fuel pumps suitable for use in the present invention provide a liquid fuel flow rate of from 10 to 50 mL/min. In some embodiments fuel pumps suitable for use in the present invention provide a liquid fuel flow rate of from 50 mL/min or greater. In these and other embodiments, mass flow controllers suitable for use in the present invention provide a flow rate of 50-2000 mL/min.

[0076] In some embodiments of desulfurization systems according to the present invention, temperature control of the liquid fuel, reactor, and desulfurized liquid fuel is provided via a temperature controller in communication with one or more thermocouples located on fluid conduits

and reactors which controls heating elements and heat exchanges present within the desulfurization system. Suitable temperature controllers are known in the art. In some embodiments, temperature controllers suitable for use in the present invention maintain reactor temperatures of from about 100° C. to 450° C.

[0077] In some embodiments of desulfurization systems according to the present invention, pressure control of the desulfurization systems is provided via a pressure controller in communication with a pressure control valve, pressure transducer, and multi-way valves in the desulfurization system. Suitable pressure transducers, pressure control valves and pressure controllers are known in the art. In some embodiments, pressure controllers suitable for use in the present invention maintain reactor pressures of from 50PSI to 200 PSI.

EXAMPLES

[0078] In order to demonstrate the practice of the present invention, the following examples have been prepared and tested. The examples should not, however, be viewed as limiting the scope of the invention. The claims will serve to define the invention.

Experimental Setup

[0079] The experimental setup included a jet fuel source and fuel pump connected to a desulfurization reactor loaded with sorbent and a cartridge heater. A nitrogen gas source was also connected to the reactor to provide an inert environment prior to flowing the jet fuel, or a nitrogen/hydrogen mixture gas source to the reactor for sorbent activation or regeneration. The product stream of the reactor was connected to a condenser where desulfurized fuel was collected. For SOFC application, the fuel stream (vapor) could directly be fed to a catalytic reformer to save energy for heating.

Desulfurization Reactor Design and Manufacturing

[0080] The reactor consists of a fuel inlet, fuel outlet, and a helical path for fuel flow, all formed as a single piece. When a sulfur-containing fuel is fed through the inlet, it comes into contact with a metal oxide sorbent material in the fuel path. A heating element is placed into the reactor core to raise and control the fuel temperature. To eliminate the condenser and to increase the energy efficiency, one or two layered double heat exchangers are added to cool down the outgoing hot fuel by the incoming cold fuel and to heat up the incoming cold fuel by the outgoing hot fuel. The reactor design and 3D printing of the reactor possess advantages including (1) eliminated multiple manufacturing steps including machining, welding, and manual assembly of various components, resulting lightweight and simplified assembly process; (2) the one-piece design prevented any possible fuel leaks from the reactor to external and internal leaks in between the helix edges and the outer walls as those leaks occur in conventional designs such as those described in U.S. Pat. No. 9,238,781, resulting in high desulfurization efficiency; and (3) eliminate a water-cooled condenser and related components, further simplified the system and reduced the weight and size of the system.

[0081] The desulfurization reactor was manufactured using an industrial metal 3D printer (M 290, EOS GmbH, Germany) and 316L grade stainless steel powder. The printer utilized a 400-W Yb-fiber laser to achieve selective

laser sintering (SLS) of the metal parts. A tubular cartridge heater (9.5 mm dia.×203 mm length, 600 W, Watlow) was placed into the reactor core to raise the fuel temperature during desulfurization. A DC power supply was used to regulate the heater temperature and four K-type thermocouples were placed at different locations of the reactor surface to keep track of the temperature. The outer diameter and length (excluding the extended portion for the electric heater) of the reactor were 38.4 and 154 mm, respectively. Compared with an improved conventional (machined) reactor with a similar overall volume, the 3D printed reactor resulted in a significant reduction in weight and increase in sorbent bed volume. The reactor weight decreased by 44% (from 1417 g to 790 g), whereas the sorbent bed volume increased by 54% (from 65 mL to 100 mL).

Jet Fuel and Sorbent

[0082] The tests were conducted using F-24 jet fuel that had an initial sulfur content of 412 ppm_w. F-24 is a NATO designation for the commercial Jet A fuel with military additives. F-24 is thus similar to the military grade JP-8 fuel, the only major difference being their freezing points. A commercial regenerable metal oxide sorbent was filled in the reactor and it was activated prior to the desulfurization tests by flowing diluted hydrogen (2 vol % in nitrogen) at 400° C. for >24 h. Nitrogen gas was used to provide an inert atmosphere in the reactor with the activated sorbent before flowing the jet fuel.

Desulfurization Procedure

[0083] A micro annular gear pump (mzr-4605, HNP Mikrosysteme, Germany) was used for supplying the jet fuel to the desulfurization reactor at a flow rate of 1-3 mL/min. The operating pressure in the reactor was controlled by a relief valve and a pressure gauge was used to monitor the pressure. A constant pressure of 120 psi (8.3 bar) was maintained in the reactor for all the tests so as to keep the jet fuel in a liquid phase for the selected range of operating temperatures (150-240° C.). The desulfurized fuel was passed through a water-cooled condenser and collected in a container. The desulfurized fuel samples (20 mL each) were collected at particular time intervals and their sulfur contents were determined using a standard ultraviolet fluorescence method (ASTM D5453).

[0084] After completing each run of the desulfurization tests, the reactor was purged and dried with nitrogen gas. The used sorbent beads were either regenerated in situ or replaced with new sorbent beads for the next run. The regeneration procedure was the same as the initial activation procedure for the as-received sorbent.

Desulfurizer Performance

[0085] Various tests were conducted using the developed reactor to elucidate the effect of variables such as operating temperature, fuel flow rate, and sorbent regeneration on the desulfurization performance. For example, one test included recording the concentration of sulfur compounds in the desulfurized fuel as a function of the amount of fuel treated at different temperatures for a constant flow rate of 2 mL/min. It was found that there was a strong influence of the reactor temperature on the sulfur content of the treated fuel. Although a higher temperature is preferable for the effective removal of sulfur under given conditions, it demands a

higher operating pressure to keep the fuel in the liquid phase as well as thicker wall to be mechanically safe for the high-temperature and high-pressure vessel. The reactor temperature was limited to 240° C. for further desulfurization tests.

[0086] The effects of residence time of the liquid fuel within the reactor was studied by varying flow rates during desulfurization tests. Liquid fuel flow rates were varied across a range of 1-3 mL/min. As expected, the flow rate of 1 mL/min resulted in the lowest output sulfur content for a given amount of fuel treated. The lower flow rate provided a longer residence time for fuel to be in contact with the sorbent beads, thus enabling better sulfur adsorption. Further testing found that after treating two liters of jet fuel, the sulfur concentration in the treated fuel was below 3 ppm_w when the fuel flow rate through the reactor was maintained at 3 mL/min.

[0087] Regeneration tests confirmed reusability of the sorbent bed after a first regeneration cycle. The regenerated sorbent materials were found to appropriately remove sulfur from the liquid fuel after the first regeneration cycle.

[0088] In light of the foregoing, it should be appreciated that the present invention significantly advances the art by providing a compact and lightweight desulfurization reactor that is structurally and functionally improved in a number of ways. While particular embodiments of the invention have been disclosed in detail herein, it should be appreciated that the invention is not limited thereto or thereby inasmuch as variations on the invention herein will be readily appreciated by those of ordinary skill in the art. The scope of the invention shall be appreciated from the claims that follow.

What is claimed is:

1. A single-piece desulfurization reactor comprising:
 - a fuel inlet,
 - a fuel outlet,
 - a helical fuel path in fluid communication with the fuel inlet and the fuel outlet, wherein the helical fuel path includes a metal oxide sorbent material,
 - a reactor core formed within the single-piece desulfurization reactor configured to externally receive a heating element, wherein the helical fuel path is formed around the reactor core.
2. The single-piece desulfurization reactor of claim 1, wherein the helical fuel path comprises:
 - a single-helical reactor portion including the metal oxide sorbent material formed around the reactor core, and one or more heat exchange portions comprising a double-helical portion formed around the single-helical portion in fluid communication with the fuel inlet, the single-helical reactor portion, and the fuel outlet.
3. The single-piece desulfurization reactor of claim 1, wherein a liquid fuel feedstock cannot bypass a portion of the helical fuel path within the single-piece desulfurization reactor.
4. The single-piece desulfurization reactor of claim 1, wherein the metal oxide sorbent material comprises nickel oxide.
5. The single-piece desulfurization reactor of claim 1, wherein the metal oxide sorbent material is activated.
6. The single-piece desulfurization reactor of claim 1, wherein the single-piece desulfurization reactor produces a treated output including 15 ppm_w sulfur or less.
7. A system for removing sulfur from a liquid fuel feedstock, the system comprising:

- a single-piece desulfurization reactor for producing a desulfurized liquid fuel stream, the single-piece desulfurization reactor comprising:
 - a fuel inlet,
 - a fuel outlet,
 - a helical fuel path in fluid communication with the fuel inlet and the fuel outlet, wherein the helical fuel path includes a metal oxide sorbent material,
 - a reactor core formed within the single piece desulfurization reactor configured to externally receive at least one heating element, wherein the helical fuel path is formed around the reactor core;
 - a heat exchanger for cooling the desulfurized liquid fuel stream in fluid communication with the fuel outlet.
- 8. The system of claim 7, wherein the helical fuel path comprises:
 - a single-helical reactor portion including the metal oxide sorbent material formed around the reactor core, and one or more heat exchange portions comprising a double-helical portion formed around the single-helical portion in fluid communication with the fuel inlet, the single-helical reactor portion, and the fuel outlet.
- 9. The system of claim 7, wherein the heat exchanger is a condenser, wherein the condenser removes heat from the desulfurized liquid fuel stream via heat exchange with a water stream, wherein after removing heat from the desulfurized liquid fuel stream a warmed water stream is produced.
- 10. The system of claim 7, wherein the heat exchanger comprises a fan and a tube-in-tube heat exchanger.
- 11. The system of claim 10, wherein the tube-in-tube heat exchanger comprises an interior tube positioned within an exterior tube, wherein the interior tube includes an interior inlet in fluid communication with the liquid fuel feedstock and an interior outlet in fluid communication with the fuel inlet of the reactor, wherein the exterior tube includes an exterior inlet in fluid communication with the fuel outlet of the reactor, and an exterior outlet in fluid communication with a desulfurized fuel reservoir.
- 12. The system of claim 11, wherein the tube-in-tube heat exchanger is configured for counter-current flow with the interior inlet and exterior outlet being collocated at a first

end of the tube-in-tube heat exchanger and the interior outlet and exterior inlet being collocated at a second end of the tube-in-tube heat exchanger.

13. The system of claim 12, wherein the system excludes any further heating source apart from the at least one heating element in the reactor core.

14. The system of claim 12, wherein the tube-in-tube heat exchanger further includes an external fan for moving air across the tube-in-tube heat exchanger.

15. The system of claim 7, further comprising an activation and regeneration system comprising a source of nitrogen gas, hydrogen gas, and mixtures thereof.

16. The system of claim 7, wherein one or more additional single-piece desulfurization reactors in fluid communication with the liquid fuel feedstock.

17. The system of claim 16, wherein the single-piece desulfurization reactors and the one or more additional single-piece desulfurization reactors are independently controlled to provide continuously produce the desulfurized fuel stream.

18. The system of claim 16, wherein the single-piece desulfurization reactor is configured for quick-replacement by one of the at least one or more additional single-piece desulfurization reactors to maintain continuous production of the desulfurized liquid fuel stream.

19. The system of claim 7, wherein the liquid fuel feedstock comprises JP-8 fuel.

20. A method of removing sulfur from a liquid fuel feedstock, the method comprising:

- providing the liquid fuel feedstock to the single-piece desulfurization reactor of claim 1, wherein the liquid fuel feedstock is in fluid communication with the fuel inlet;
- flowing the liquid fuel feed stock through the helical fuel path to contact the metal oxide sorbent material;
- controlling a temperature of a heating element located with the reactor core;
- controlling a pressure within the single-piece desulfurization reactor; and
- collecting a desulfurized liquid fuel stream produced by the single-piece desulfurization reactor from the fuel outlet.

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